

# CO2\_AT2 — Visualization Development Exercise

Topics: Directory of Visualizations · Visualizing Amounts · Visualizing Distributions

## QUESTION-1

### 1. Comparing the population of 8 countries in the current year.

**Chart Type:** Bar Chart (Amounts)

**Justification:**

A bar chart is the best choice because it allows easy comparison of population values across multiple countries. The height or length of the bars makes it simple to identify the countries with the highest and lowest populations.

### 2. Showing how delivery times vary across 5 courier companies.

**Chart Type:** Box Plot (Distribution)

**Justification:**

A box plot effectively displays the distribution of delivery times, including the median, quartiles, spread, and any outliers. It helps compare the consistency and variability of delivery performance among courier companies.

### 3. Showing the proportion of a household budget spent on rent, food, transport, and savings.

**Chart Type:** Pie Chart (Proportion)

**Justification:**

A pie chart is suitable because it represents each expense as a percentage of the total household budget. It clearly illustrates how the total budget is divided among different spending categories.

### 4. Showing the relationship between hours studied and exam marks for 50 students.

**Chart Type:** Scatter Plot (X–Y Relationship)

**Justification:**

A scatter plot is ideal for analyzing the relationship between two continuous variables. It helps identify trends, correlations, and patterns between study hours and exam performance.

**5. Showing average temperature for each month across 3 different cities at once.**

**Chart Type:** Line Chart (Temporal)

**Justification:**

A line chart is the best option because it shows how temperature changes over time. Multiple lines allow easy comparison of monthly temperature trends across the three cities.

**6. Showing which airports are directly connected by flight routes.**

**Chart Type:** Network Diagram (Network)

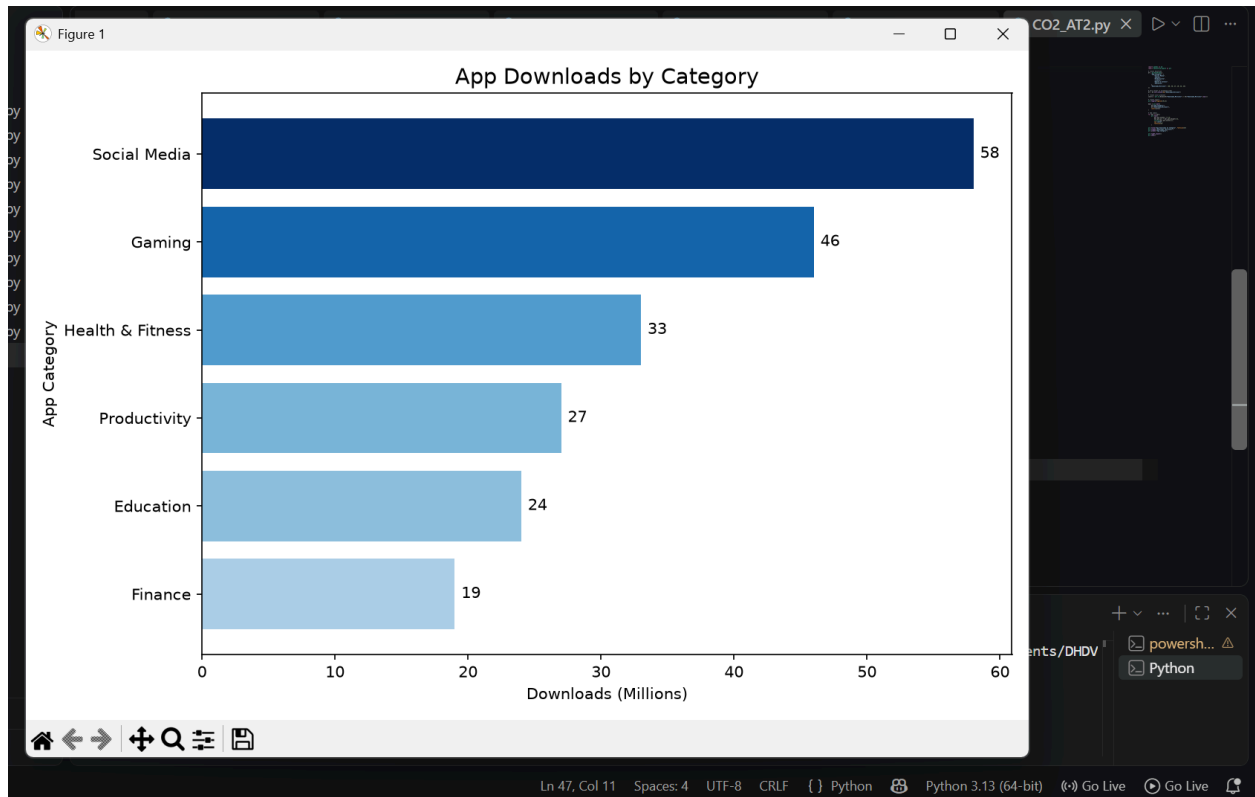
**Justification:**

A network diagram is specifically designed to represent connections between entities. Airports are shown as nodes and flight routes as links, making it easy to visualize direct connections.

**Final Summary Table**

| Scenario                                | Best Chart Type | Visualization Category |
|-----------------------------------------|-----------------|------------------------|
| Population of 8 countries               | Bar Chart       | Amounts                |
| Delivery times across courier companies | Box Plot        | Distribution           |
| Household budget allocation             | Pie Chart       | Proportion             |
| Hours studied vs. Exam marks            | Scatter Plot    | X–Y Relationship       |
| Monthly temperature of 3 cities         | Line Chart      | Temporal               |
| Airport flight connections              | Network Diagram | Network                |

## QUESTION-2



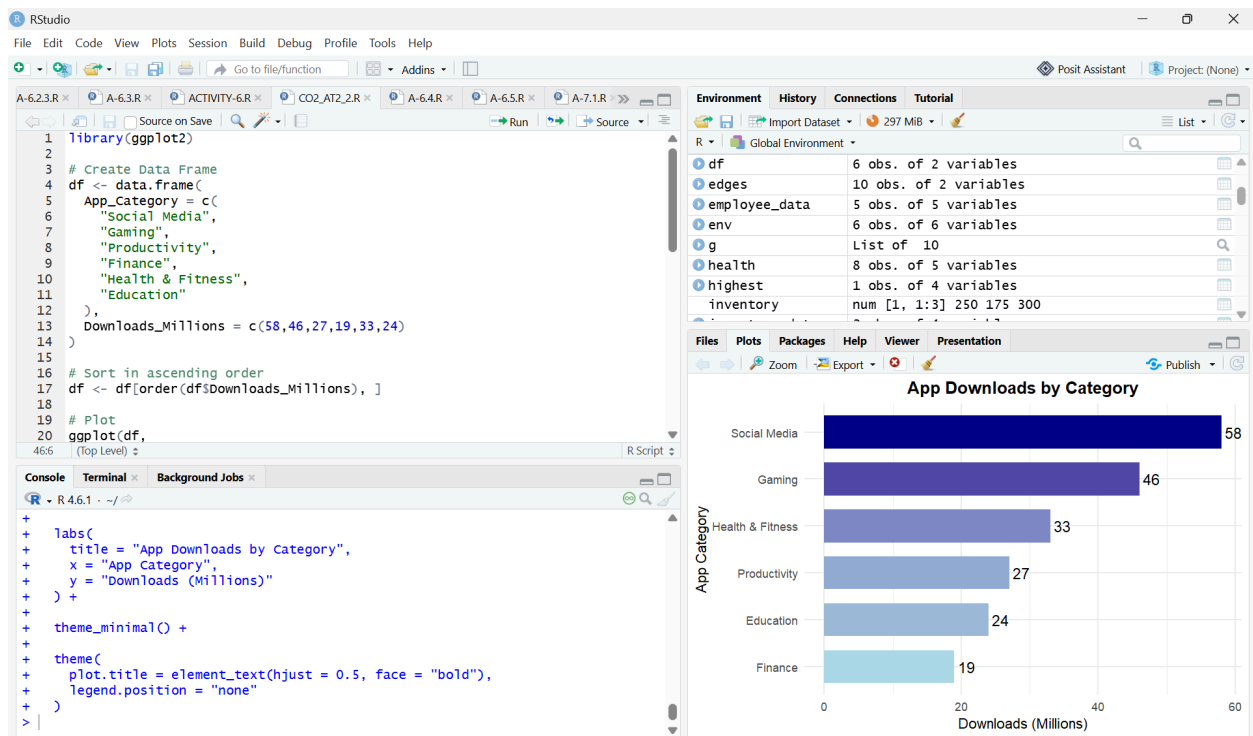
### CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
# Create DataFrame
df = pd.DataFrame({
    "App_Category": [
        "Social Media",
        "Gaming",
        "Productivity",
        "Finance",
        "Health & Fitness",
        "Education"
    ],
    "Downloads_Millions": [58, 46, 27, 19, 33, 24]
})
# Sort values in ascending order
df = df.sort_values(by="Downloads_Millions")
# Create color gradient
colors = plt.cm.Blues(df["Downloads_Millions"] / df["Downloads_Millions"].max())
# Create figure
plt.figure(figsize=(9,6))
```

```

bars = plt.barh(
  df["App_Category"],
  df["Downloads_Millions"],
  color=colors
)
# Add labels
for bar in bars:
  plt.text(
    bar.get_width() + 0.5,
    bar.get_y() + bar.get_height()/2,
    str(int(bar.get_width())),
    va='center',
    fontsize=10
  )
plt.title("App Downloads by Category", fontsize=14)
plt.xlabel("Downloads (Millions)")
plt.ylabel("App Category")
plt.tight_layout()
plt.show()

```



## CODE:

```

library(ggplot2)
# Create Data Frame
df <- data.frame(

```

```

App_Category = c(
  "Social Media",
  "Gaming",
  "Productivity",
  "Finance",
  "Health & Fitness",
  "Education"
),
Downloads_Millions = c(58,46,27,19,33,24)
)
# Sort in ascending order
df <- df[order(df$Downloads_Millions), ]
# Plot
ggplot(df,
  aes(
    x = reorder(App_Category, Downloads_Millions),
    y = Downloads_Millions,
    fill = Downloads_Millions
  )) +

  geom_col(width = 0.7) +
  coord_flip() +
  geom_text(
    aes(label = Downloads_Millions),
    hjust = -0.2,
    size = 4
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "App Downloads by Category",
    x = "App Category",
    y = "Downloads (Millions)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "none"
  )

```

## Expected Output

A **horizontal bar chart** where:

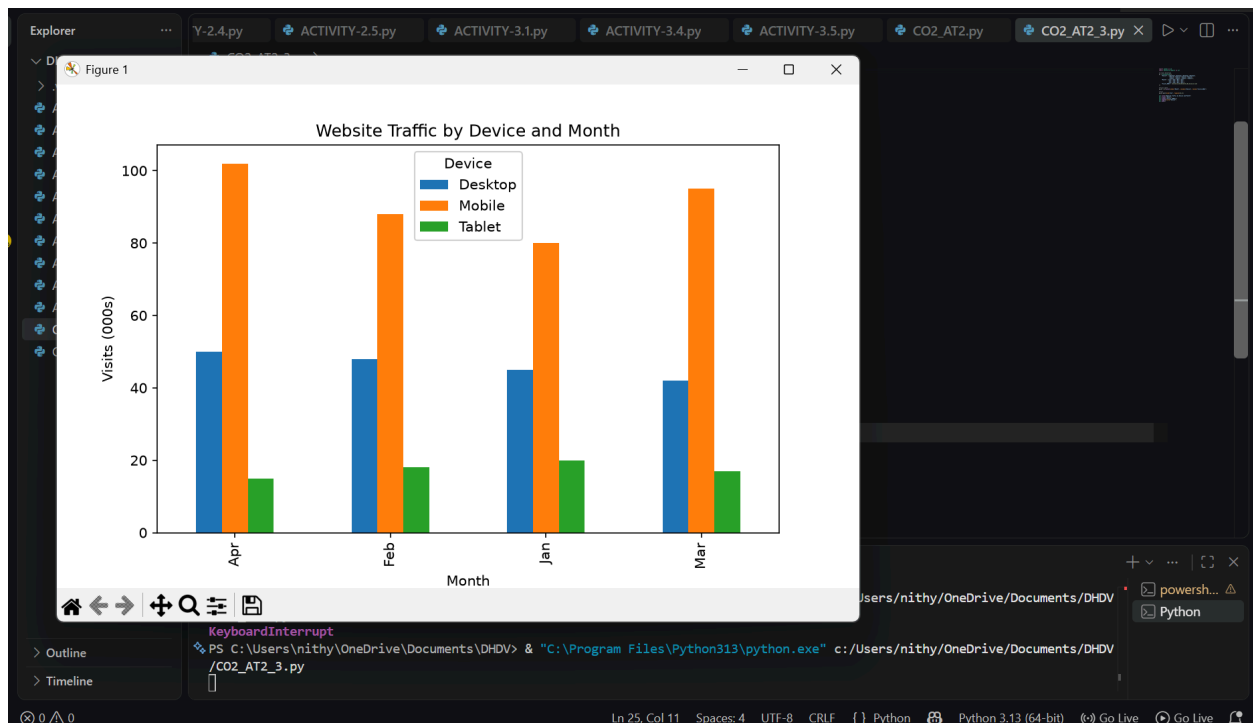
- Finance appears first (19 million)
- Education (24 million)
- Productivity (27 million)
- Health & Fitness (33 million)
- Gaming (46 million)
- Social Media (58 million)

Bars become **darker** as downloads increase, and each bar displays its exact value.

## Interpretation

- **Social Media** has the highest downloads (**58 million**), indicating it is the most popular app category.
- **Gaming** is the second most downloaded category with **46 million** downloads.
- **Health & Fitness** has moderate popularity with **33 million** downloads.
- **Finance** records the lowest downloads (**19 million**).
- The color gradient makes it easy to identify categories with higher download counts at a glance.

## QUESTION-3



## CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

# Create DataFrame

df = pd.DataFrame({

    "Device": ["Desktop", "Desktop", "Desktop", "Desktop",

               "Mobile", "Mobile", "Mobile", "Mobile",

               "Tablet", "Tablet", "Tablet", "Tablet"],

    "Month": ["Jan", "Feb", "Mar", "Apr",

              "Jan", "Feb", "Mar", "Apr",

              "Jan", "Feb", "Mar", "Apr"],

    "Visits_000s": [45, 48, 42, 50, 80, 88, 95, 102, 20, 18, 17, 15]

})

# Pivot table

pivot = df.pivot(index="Month", columns="Device", values="Visits_000s")

# Plot

pivot.plot(kind="bar", figsize=(8,5))

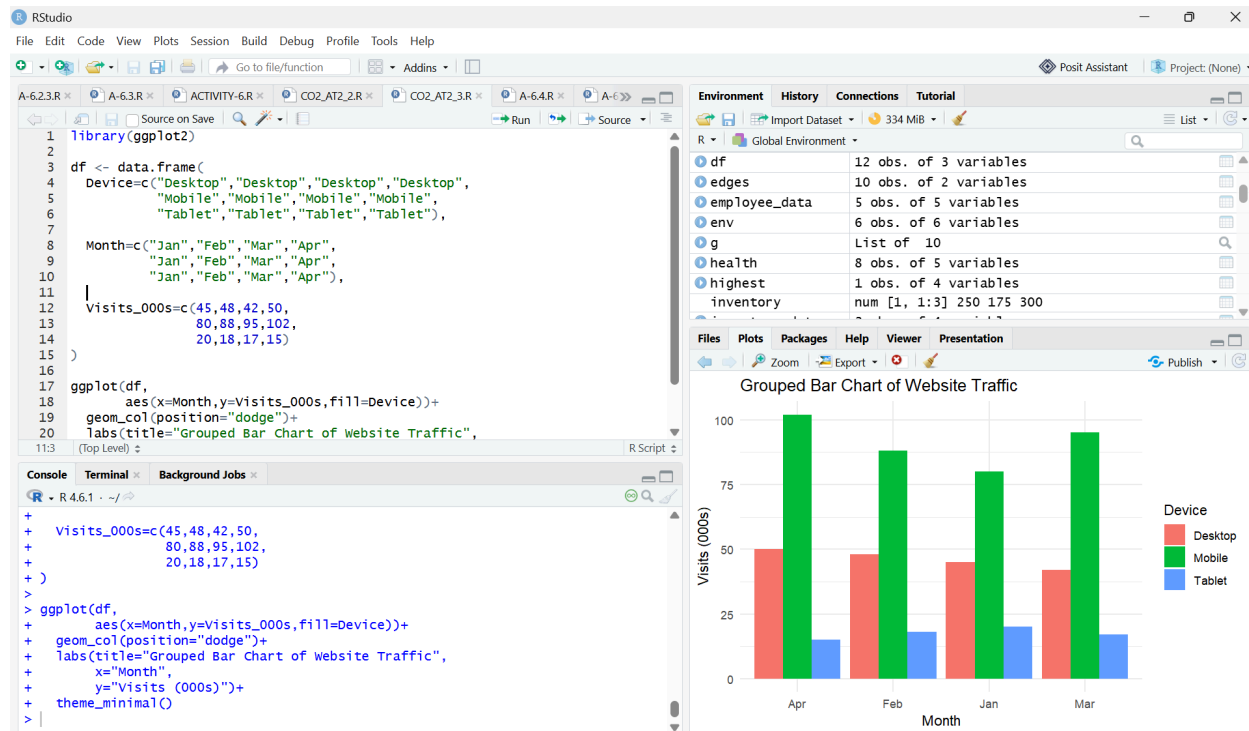
plt.title("Website Traffic by Device and Month")

plt.xlabel("Month")

plt.ylabel("Visits (000s)")

plt.legend(title="Device")

plt.show()
```



CODE:

```

library(ggplot2)
df <- data.frame(
  Device=c("Desktop","Desktop","Desktop","Desktop",
           "Mobile","Mobile","Mobile","Mobile",
           "Tablet","Tablet","Tablet","Tablet"),
  Month=c("Jan","Feb","Mar","Apr",
          "Jan","Feb","Mar","Apr",
          "Jan","Feb","Mar","Apr"),
  Visits_000s=c(45,48,42,50,
                80,88,95,102,
                20,18,17,15)
)
ggplot(df,
  aes(x=Month,y=Visits_000s,fill=Device))+
  geom_col(position="dodge")+
  labs(title="Grouped Bar Chart of Website Traffic",
       x="Month",
       y="Visits (000s)")+
  theme_minimal()

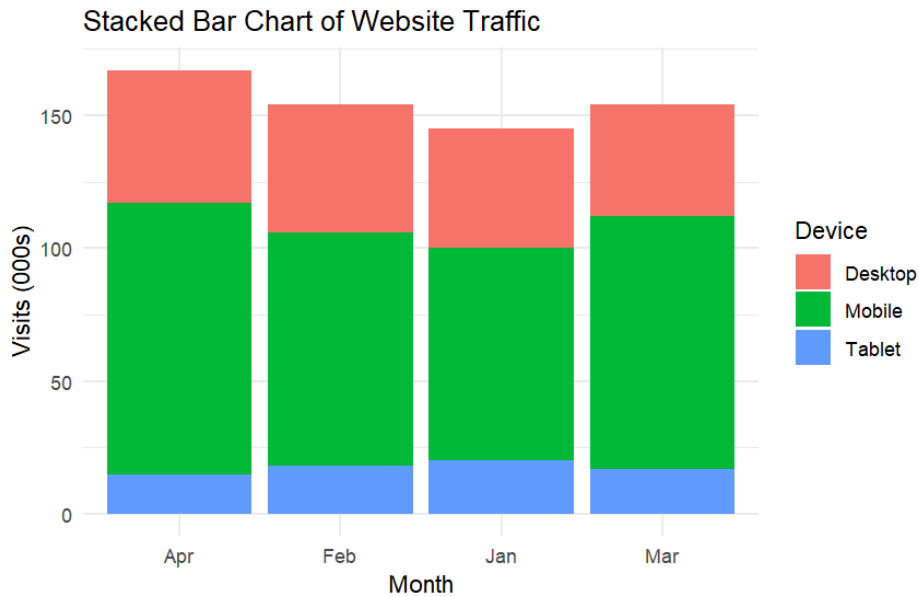
```



## Interpretation

- Mobile traffic is highest in every month.
- Desktop traffic remains moderate.
- Tablet traffic is the lowest.

## Task 8: Stacked Bar Chart



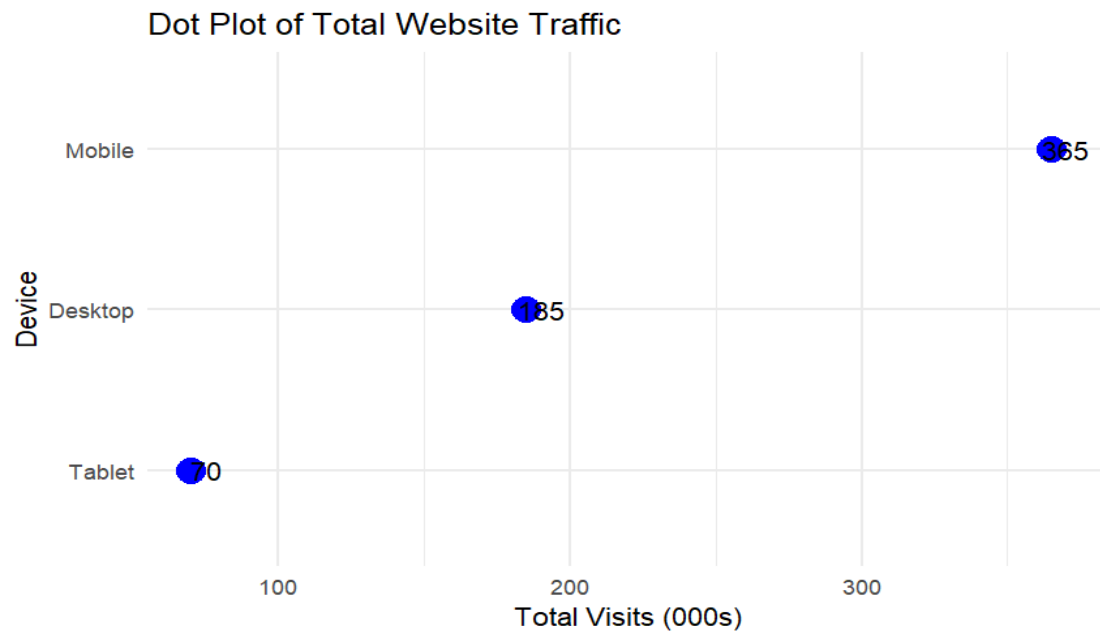
CODE:

```
ggplot(df,  
aes(x=Month,  
y=Visits_000s,  
fill=Device))+  
geom_col(position="stack")+  
labs(title="Stacked Bar Chart of Website Traffic",  
x="Month",  
y="Visits (000s)")+  
theme_minimal()
```

## Interpretation

- April records the highest overall website traffic.
- Mobile contributes the largest share every month.
- Tablet contributes the least.

## Task 9: Dot Plot



CODE:

```
library(dplyr)

total <- df %>%

group_by(Device) %>%

summarise(Total=sum(Visits_000s)) %>%

arrange(desc(Total))

ggplot(total,

aes(x=Total,

y=reorder(Device,Total)))+

geom_point(size=5,color="blue")+
```

```
geom_text(aes(label=Total),
nudge_x=5)+
labs(title="Dot Plot of Total Website Traffic",
x="Total Visits (000s)",
y="Device")+
theme_minimal()
```

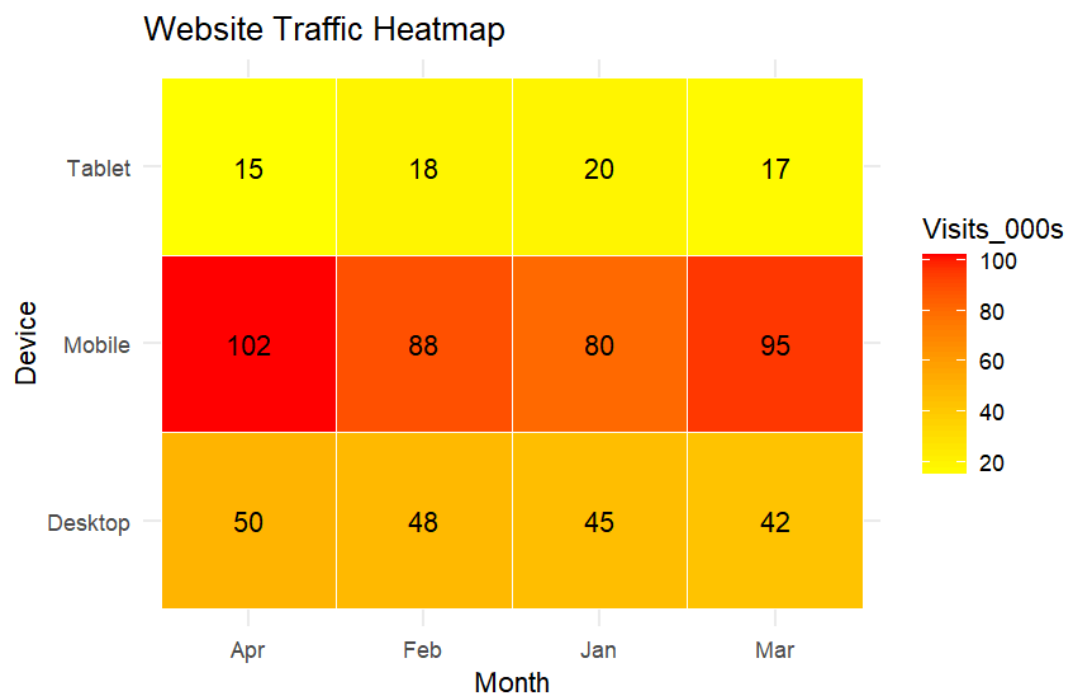
## Interpretation

Totals:

- Mobile = **365**
- Desktop = **185**
- Tablet = **70**

Mobile receives the highest traffic, while Tablet receives the lowest.

## Task 10: Heatmap



**CODE:**

```
library(ggplot2)

ggplot(df,

aes(x=Month,

y=Device,

fill=Visits_000s))+

geom_tile(color="white")+

geom_text(aes(label=Visits_000s),

color="black")+

scale_fill_gradient(

low="yellow",

high="red"

)+

labs(title="Website Traffic Heatmap",

x="Month",

y="Device")+

theme_minimal()
```

**Expected Outputs****1. Grouped Bar Chart**

- Compares Desktop, Mobile, and Tablet traffic side-by-side for each month.

**2. Stacked Bar Chart**

- Shows total monthly traffic with each device's contribution stacked.

**3. Dot Plot**

- Displays total traffic for each device ranked from highest to lowest.

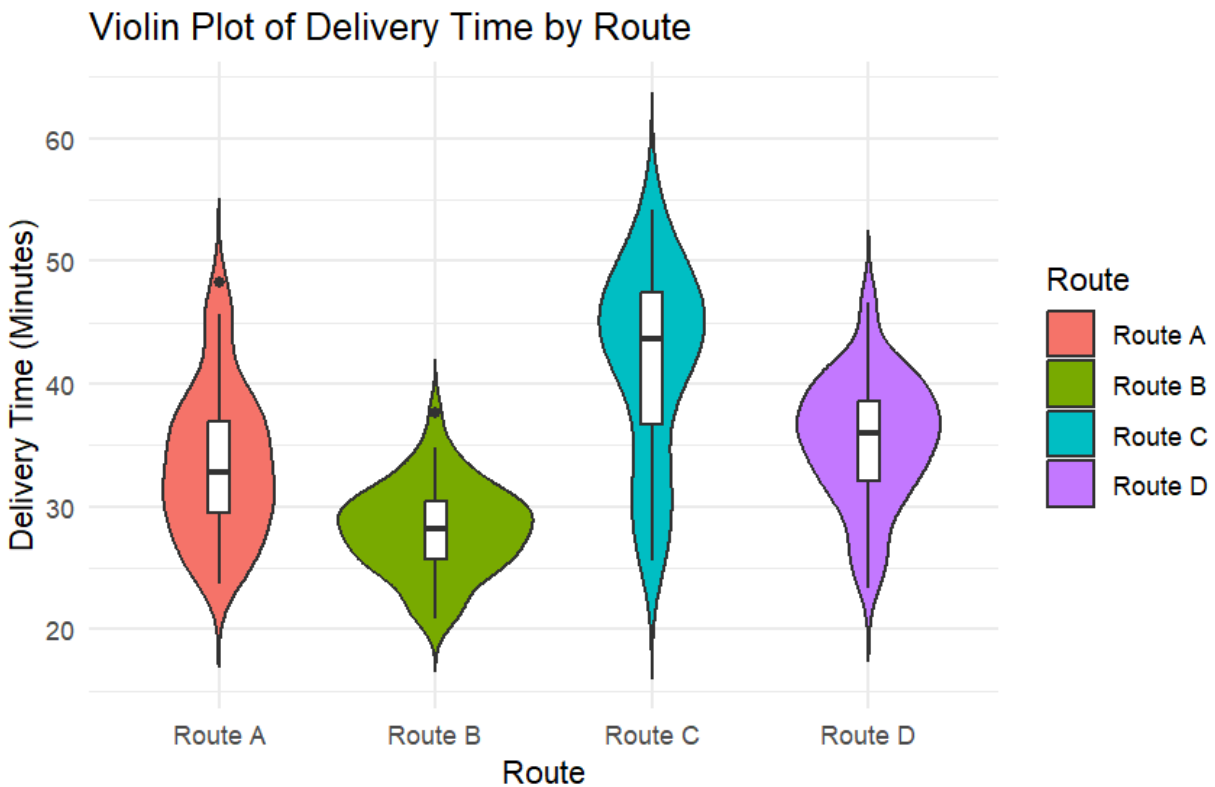
#### 4. Heatmap

- Uses color intensity to indicate traffic volume across devices and months.

#### Overall Interpretation

- **Mobile** consistently has the highest website traffic, increasing from **80,000 visits in January** to **102,000 visits in April**.
- **Desktop** traffic remains relatively stable between **42,000 and 50,000 visits**.
- **Tablet** traffic declines steadily from **20,000 visits in January** to **15,000 visits in April**.
- **April** has the highest overall website traffic, largely driven by increased mobile usage.
- The heatmap clearly highlights **Mobile in April** as the highest traffic combination, while **Tablet in April** shows the lowest traffic.

#### QUESTION-4



**CODE:**

```
library(ggplot2)

# Set seed

set.seed(7)

# Generate data

routeA <- rnorm(50,32,6)

routeB <- rnorm(50,28,4)

routeC <- rnorm(50,40,9)

routeD <- rnorm(50,35,5)

# Data Frame

df <- data.frame(

  Route=c(rep("Route A",50),

           rep("Route B",50),

           rep("Route C",50),

           rep("Route D",50)),

  Delivery_Time=c(routeA,

                   routeB,

                   routeC,

                   routeD))

# Violin Plot

ggplot(df,

aes(x=Route,

y=Delivery_Time,
```

```

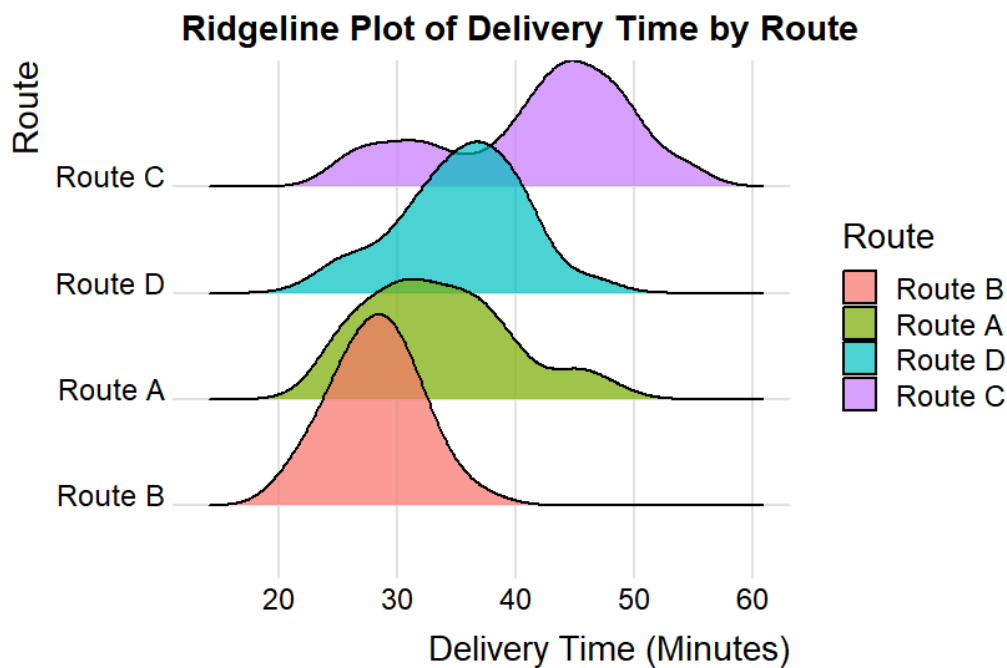
fill=Route)))+
geom_violin(trim=FALSE)+
geom_boxplot(width=0.1,
fill="white")+
labs(title="Violin Plot of Delivery Time by Route",
x="Route",
y="Delivery Time (Minutes)")+
theme_minimal()

```

### Interpretation (Violin Plot)

- **Route B** has the lowest median delivery time, making it the fastest route.
- **Route C** has the widest distribution, indicating the greatest variability in delivery times.
- **Route A** and **Route D** have moderate delivery times with less variation than Route C.

### Task 12 – Ridgeline Plot



## Expected Output

### Violin Plot

- Four violin plots representing Route A, Route B, Route C, and Route D.
- Each violin contains a boxplot showing the median and quartiles.

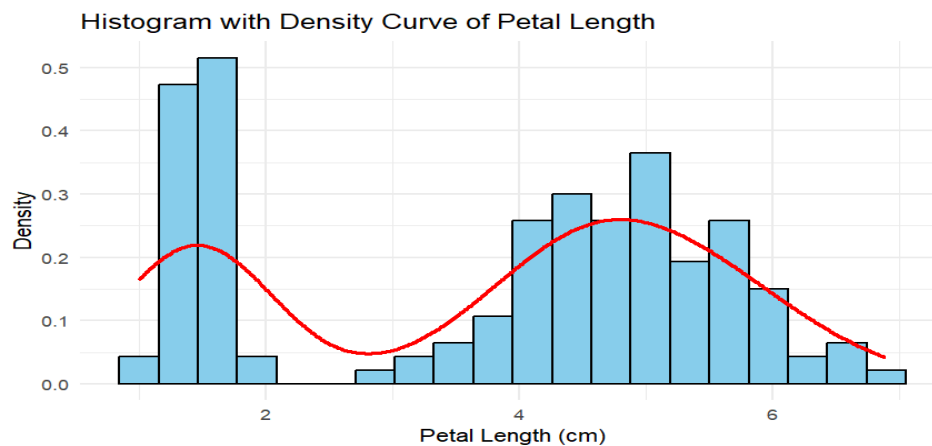
### Ridgeline Plot

- Four overlapping density curves.
- Routes arranged from fastest (lowest mean delivery time) at the bottom to slowest (highest mean delivery time) at the top.

### Overall Interpretation

- Route B has the shortest average delivery time, making it the most efficient route.
- Route C has the highest average delivery time and the greatest spread, indicating more variability and inconsistency.
- Route A and Route D show moderate delivery times with relatively balanced distributions.
- The violin plot highlights the spread and median of each route, while the ridgeline plot makes it easy to compare the distribution shapes across all routes.

## QUESTION-5





## CODE:

```
# Load library

library(ggplot2)

# Histogram with Density Curve

ggplot(iris,

       aes(x = Petal.Length)) +

  geom_histogram(

    aes(y = after_stat(density)),

    bins = 20,

    fill = "skyblue",

    color = "black"

  ) +

  geom_density(

    color = "red",

    linewidth = 1

  ) +

  labs(

    title = "Histogram with Density Curve of Petal Length",

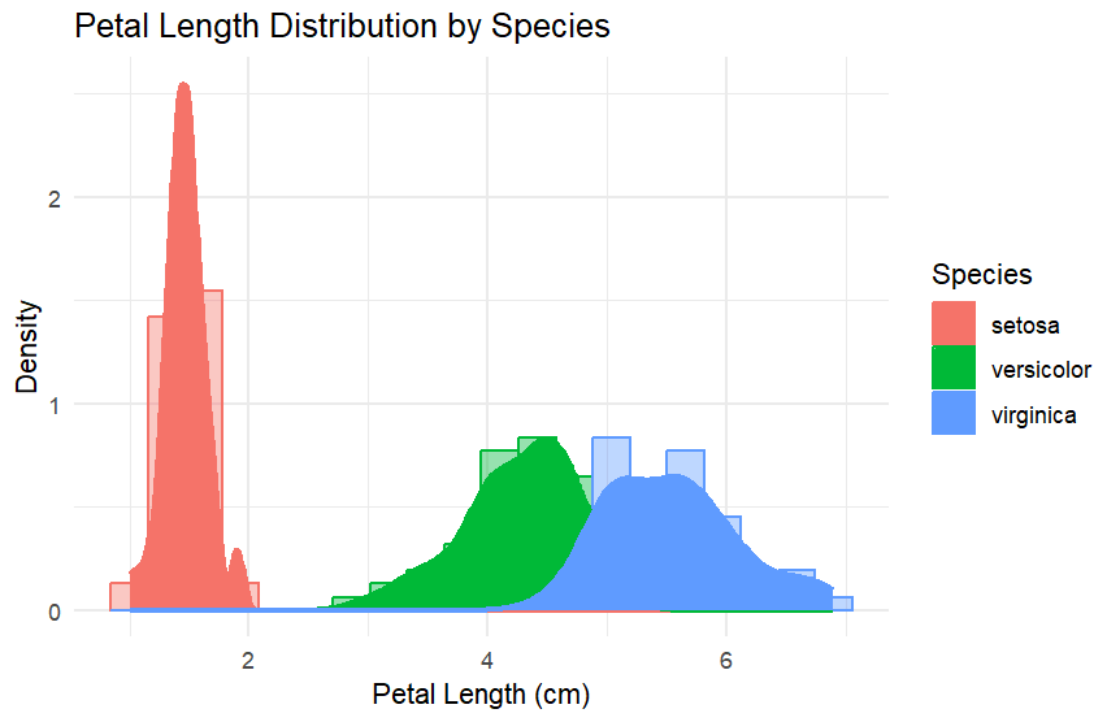
    x = "Petal Length (cm)",

    y = "Density"

  ) +

  theme_minimal()
```

## Task 14 – Histogram & KDE Split by Species



### Expected Output

#### 1. Histogram with KDE (Overall)

- A histogram of Petal Length with 20 bins.
- A smooth KDE curve overlaid on the histogram to show the overall distribution.

#### 2. Histogram with KDE by Species

- Three overlapping histograms and KDE curves for:
  - Setosa
  - Versicolor
  - Virginica
- Different colors distinguish the species and allow comparison of their distributions.

### Interpretation

#### Overall Histogram

- Petal Length values range approximately from 1 cm to 7 cm.
- The KDE curve shows the overall distribution of petal lengths.

- The distribution is not perfectly symmetric because it combines measurements from three different species.

## **Grouped Histogram**

- Setosa has the smallest petal lengths and forms a distinct distribution.
- Versicolor has intermediate petal lengths with moderate spread.
- Virginica has the largest petal lengths and shows the widest range.
- The grouped histogram and KDE curves make it easy to compare the distribution shape and overlap among the three species.